Suryachalam V M

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Profile

Full-Stack and Product Engineer with hands-on production experience designing scalable distributed platforms, real-time systems, and offline-first mobile applications. Currently interning at SOCIO (campus event ecosystem), engineering Valkey distributed caching, push notification infrastructure, and hybrid Capacitor Android delivery. Vertical Head for DevZen at CHRIST University CS Club and Sub Head for INTERFACE 2026, leading technical workshops, hackathons, and cross-team developer outreach. Pursuing BSc Computer Science & Mathematics at CHRIST University.

Education

Christ University, Bengaluru 2025 – 2029
BSc Computer Science and Mathematics – Coursework: DSA, Graph Theory, Linear Algebra, DBMS, OOP

Experience

Product Engineering Intern Apr 2026 – Present
SOCIO – Campus Event Platform, CHRIST Incubation and Consultancy Foundation (CICF) *Bengaluru, India*

- Implemented a Valkey (Redis-compatible) caching layer for session management, notification queuing, and real-time attendance counters, reducing PostgreSQL query overhead by 40%
- Architected push notification infrastructure across OneSignal, FCM, Web Push/VAPID, and Service Workers, delivering 99.8% delivery rate across PWA and Native Android
- Developed an offline-first QR ticket scanner using IndexedDB/Dexie.js with background sync, eliminating duplicate scans via cooldown-based conflict resolution
- Shipped cross-platform features across Web, PWA, and Native Android (Capacitor) simultaneously in fast-paced 50+ hrs/week agile sprint cycles

Projects

SOCIO V2 – Campus Event & Operations Platform | *Next.js 15, Express 5, Supabase, Valkey, PostgreSQL* socio.christuniversity.in
- Architected monorepo automating QR ticketing, attendance tracking, and 9+ role-gated operational workflows (HOD, Dean, CFO, Volunteer, Catering) for high-traffic university events
- Implemented real-time analytical dashboards using Recharts and secure JWT session hydration on the server side

SOCIO Mobile – Hybrid Android & PWA App | *Next.js 16, Capacitor, IndexedDB, WebRTC, OneSignal* app.withsocio.com
- Engineered 22-route hybrid app with native camera/torch plugins, offline ticket caching via IndexedDB, and accelerometer-bound shake-to-scan gesture triggers
- Integrated dual-channel push alerts with Web Push VAPID and OneSignal targeting active student portals

Labyrinth – Official CS Club Platform | *Next.js 16, TypeScript, Supabase PostgreSQL, RLS, JWT/JOSE* labyrinth.christuniversity.in
- Built full-stack platform with dynamic form builder, applicant evaluation pipeline, and role-aware admin controls
- Engineered hybrid dual-storage engine with auto-failover between Supabase and local JSON, secured by JWT (JOSE) sessions and Edge Middleware route protection
- Engineered client-side .docx activity report generator with geotagged photo embedding and Excel/PDF export

FacultyApp – Institutional Workflow Management | *Flutter, Riverpod, Next.js, Prisma, Supabase* github.com/sgk18/Facultyapp
- Full-stack platform with JWT auth, Google OAuth, and granular role-based ADMIN/HOD/FACULTY access control
- Integrated Google Calendar and Gmail APIs, FCM push alerts, and Resend transactional email notifications

Technical Skills

Languages: TypeScript, JavaScript, Python, Java, Dart, SQL, PL/SQL, C++
Frontend & Mobile: React, Next.js, Flutter, Capacitor, Tailwind CSS, Framer Motion, PWA, Service Workers
Backend & Infra: Node.js, Express, PostgreSQL, Supabase (RLS), Prisma, MongoDB, Neo4j, Valkey, MySQL HeatWave, Docker, AWS, OCI
Integrations & Tools: CI/CD, FCM, OneSignal, Google OAuth, Gmail API, Resend, JWT/JOSE, Git & GitHub, Vercel, Figma

Achievements & Certifications

Oracle Data Platform 2025 Foundations Associate – Oracle University (ID: 328494602OCI25DCFA)
Certificate of Participation in First Commit – WeMakeDevs (Credential ID: J8KH4-2TG34, AWS)
Runner-Up – Frontend Frenzy, Xactitude IT Fest 2026 · **1st Place** – Mathematics Premier League (MPL), SEQUENCE 2026
3rd Prize – National Science Day Exhibition, CHRIST University · **Finisher** – 24-Hour Solo Hackathon, CHRIST × Pod.ai
Certifications: Graph Theory (CHRIST) · HackerRank Python Basic · HackerRank SQL Basic · C Programming · AI Prompting